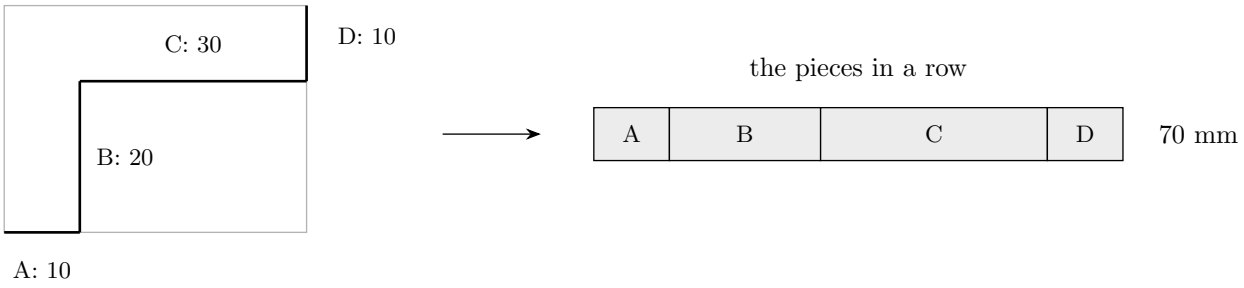
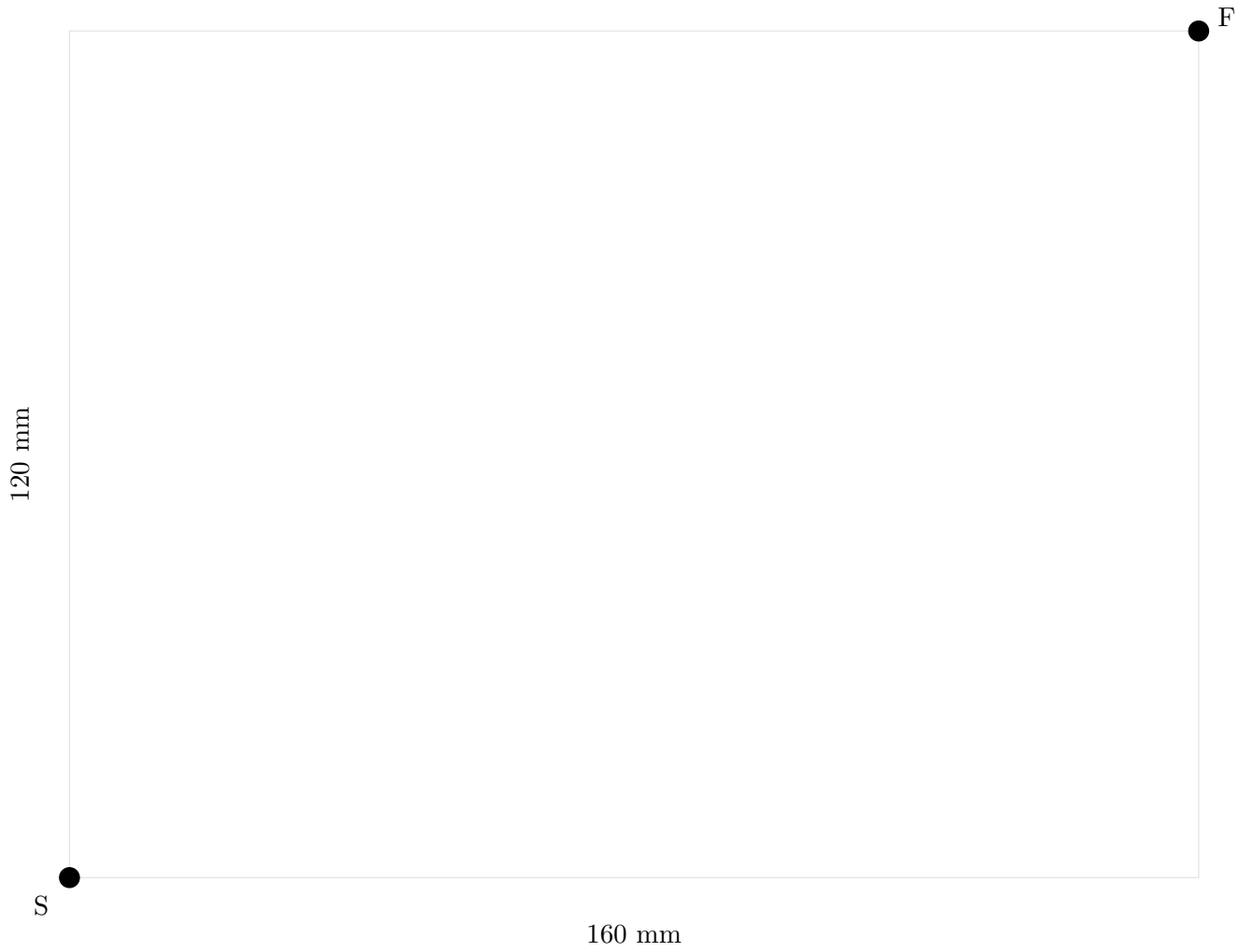
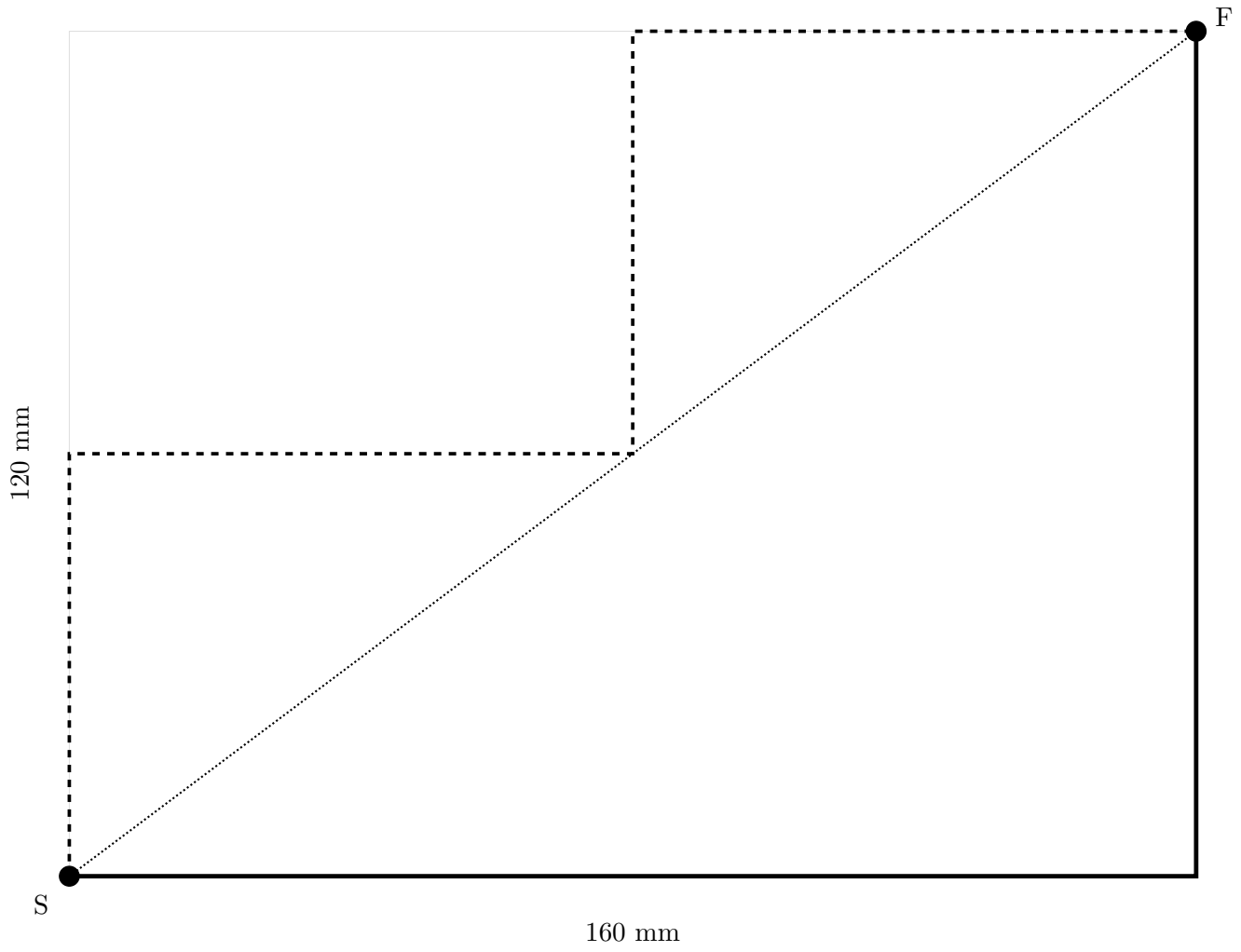
A staircase goes only right and up. Measure along all its pieces.



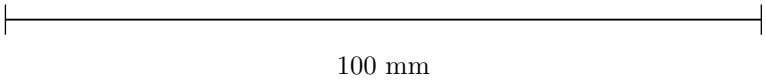
Problem 1: Make two different staircases from S to F. Use string to compare their lengths.



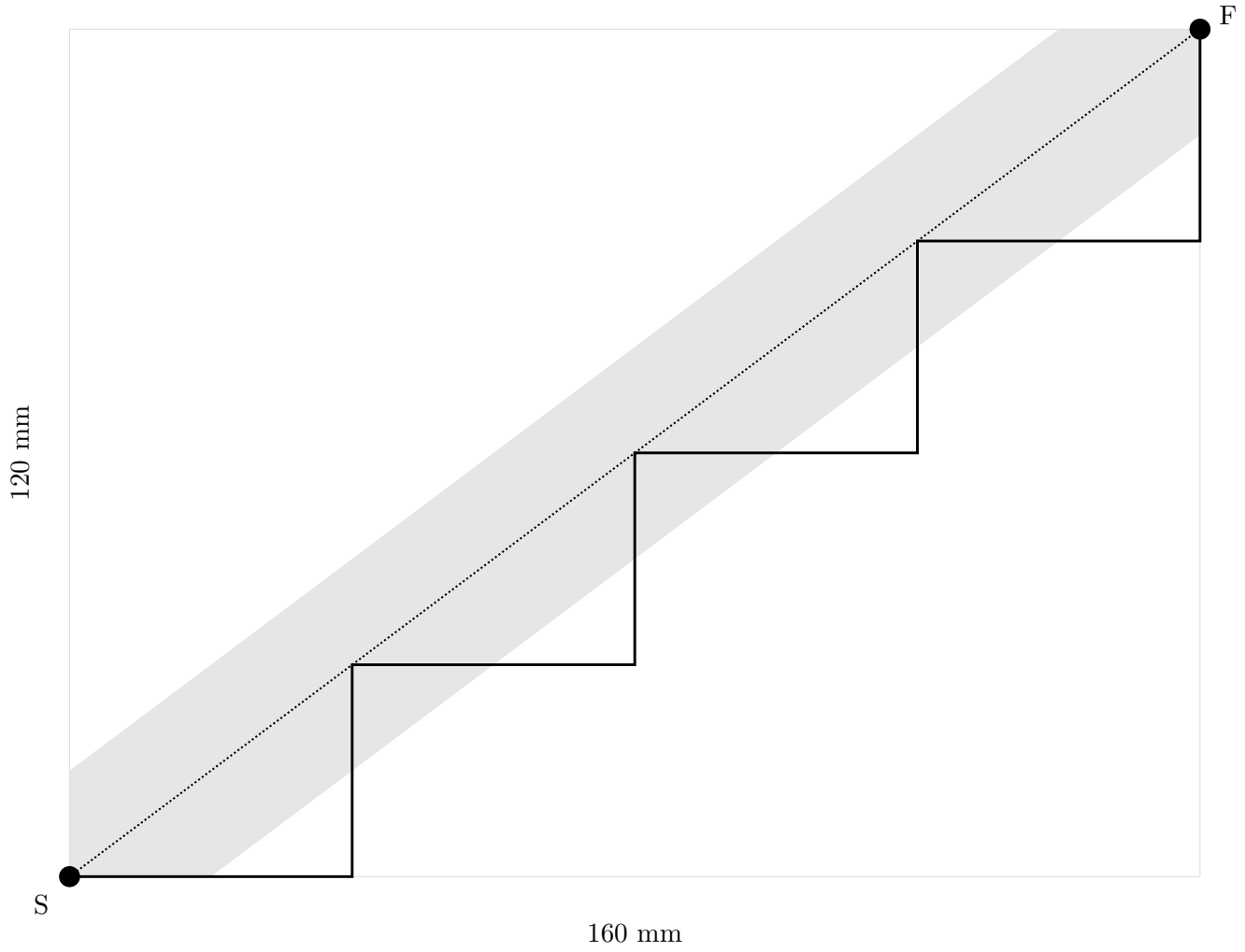
Problem 2: Compare the lengths of these two staircases. Find the exact length of each staircase.



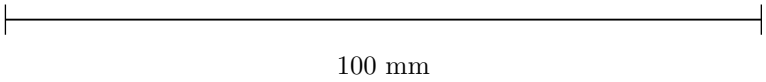
Solid: _____ mm Dashed: _____ mm



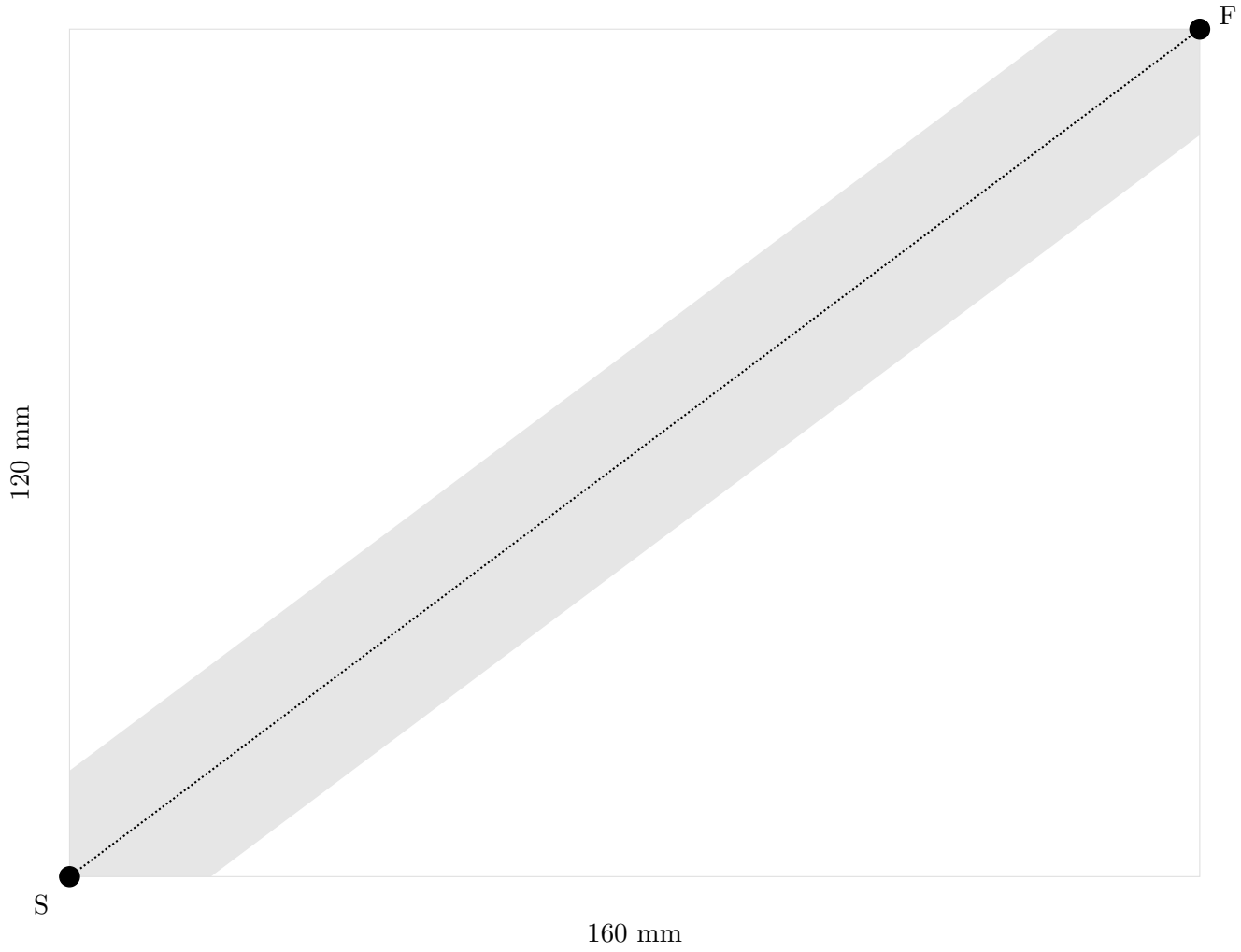
Problem 3: Make a staircase inside the gray strip. Compare the two staircase lengths with the diagonal's length.



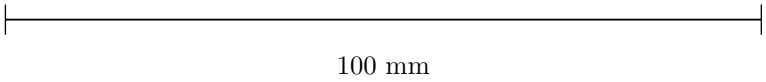
The straight diagonal is 200 mm long. The strip reaches 15 mm vertically above and below it.



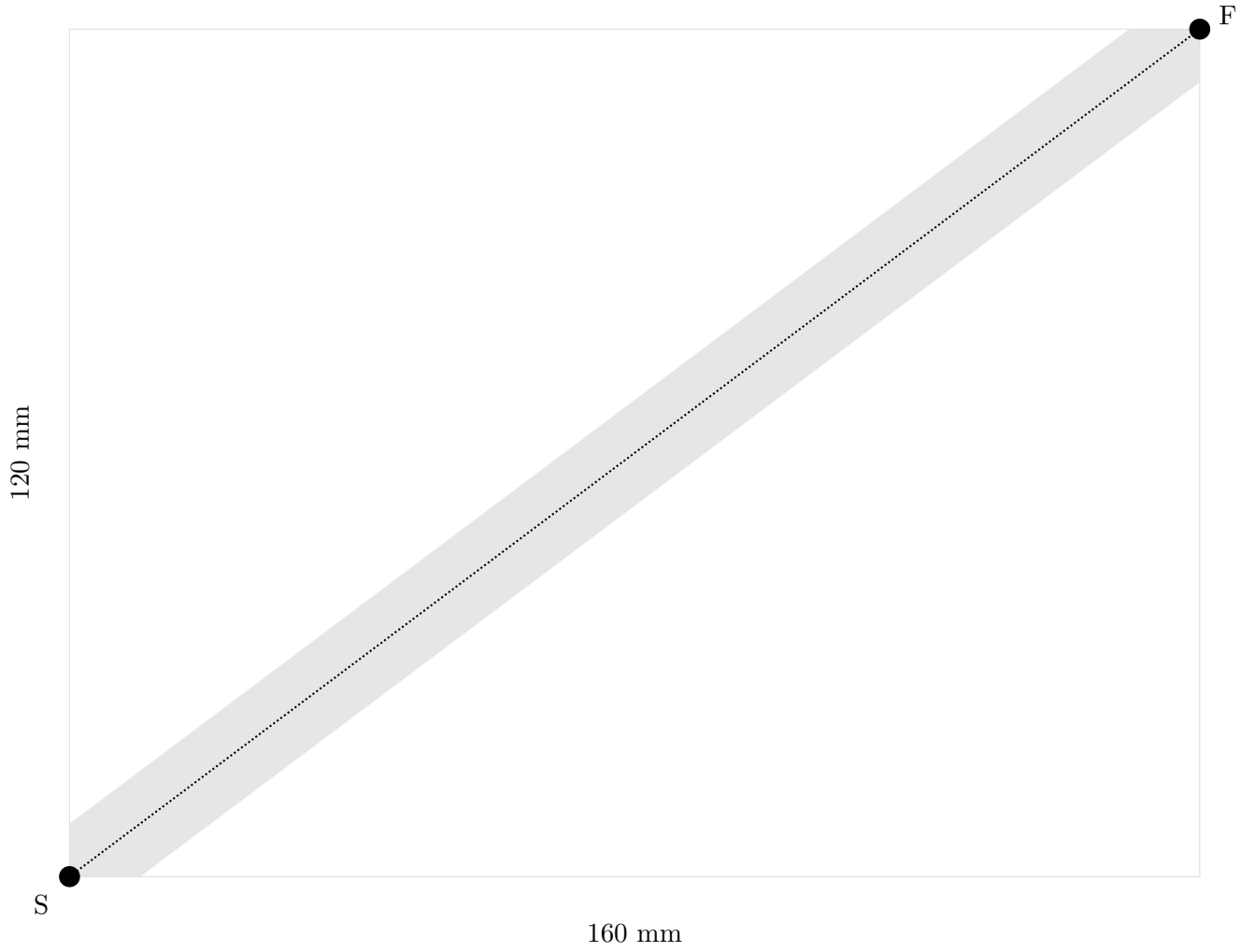
Problem 4: Find the fewest turns possible for an S-to-F staircase inside the gray strip. Its pieces may follow the strip's boundary. Explain why fewer turns cannot work.



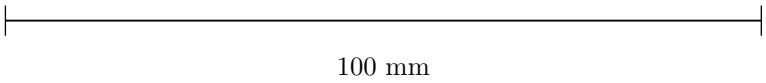
The gray strip reaches 15 mm vertically above and below the dotted diagonal.



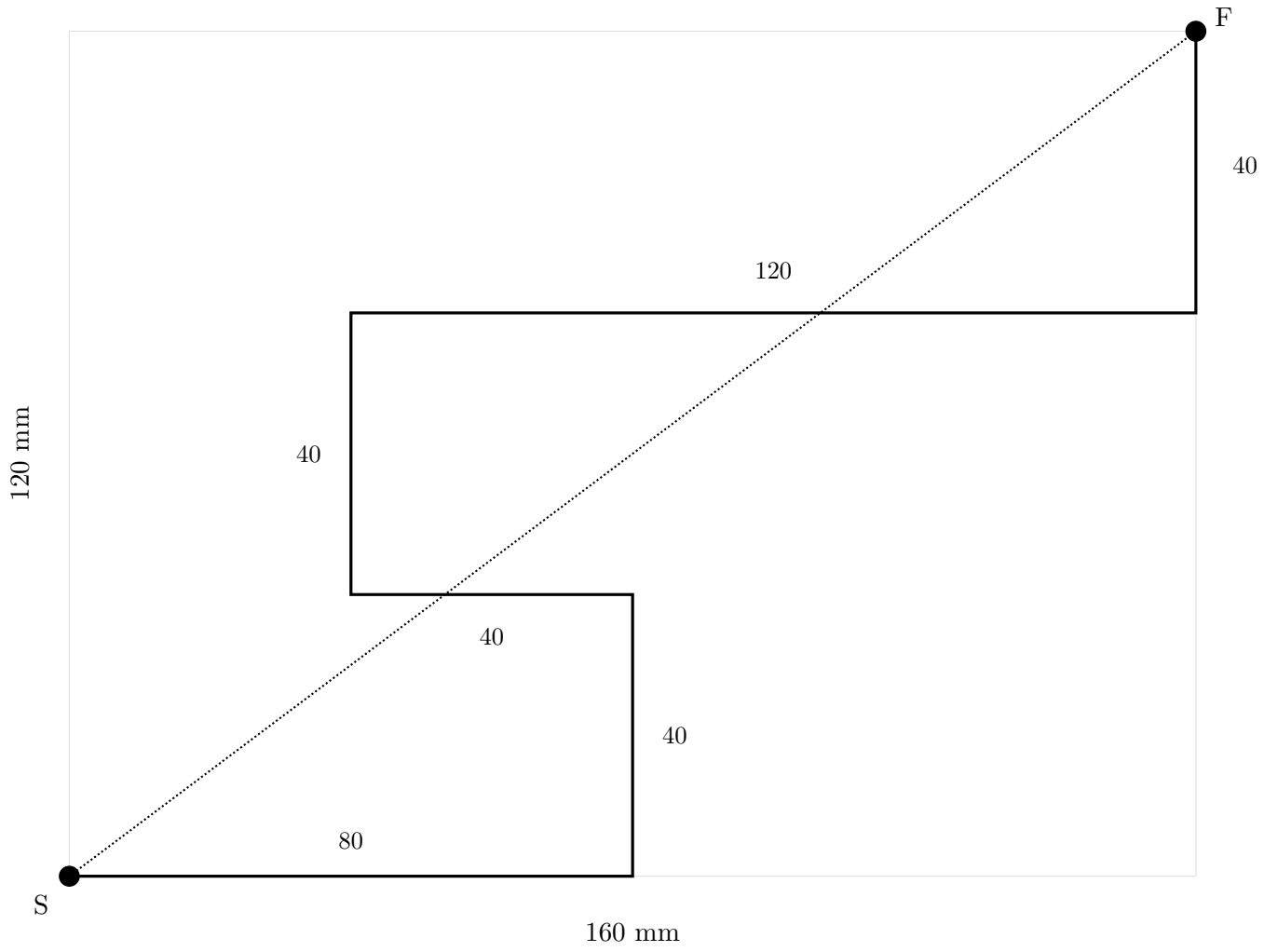
Problem 5: Make a staircase inside this narrower strip with as few turns as you can. Is it shorter than your staircase in the wider strip? Explain your answer.



The strip reaches 7.5 mm vertically above and below the diagonal.



Problem 6: Find this path's exact length. Explain which part of your argument about right-and-up staircases no longer applies.



Problem 7: A staircase may use more and smaller right-and-up pieces. Could any finite number of pieces give the same length as the 200 mm diagonal? Explain your answer.